
Diversity You Cannot See From the Mean: Calibrated Detection of Latent Failure-Mode Structure in Open-Model Ecosystems

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Standard mean-level diversity statistics for model panels — double-fault, pairwise
2 disagreement, observed agreement — are computed the same way whether the
3 panel is meant to be diverse or not: count how often members fail, or answer, alike.
4 We show this class of statistic is provably degenerate for the failure case: mean
5 pairwise co-failure is a deterministic function of item difficulty alone and carries
6 no information about shared behavior once that is accounted for. On 1,228–1,362
7 open language models across five benchmarks, the mean-level “excess” co-failure
8 a naive computation reports is identically zero under the correct null, yet a genuine,
9 calibration-only-visible residual correlation of $2.8\text{--}11.8\times$ its null survives,
10 concentrated in 4–12 latent behavioral dimensions depending on benchmark. An
11 uncalibrated version of the same summary statistic reads 1.5–3.6 on every bench-
12 mark before conditioning — easily misread as near-total diversity — and only
13 calibration against an exact, fit-free reference distribution reveals the real structure
14 underneath. We argue this measurement-validity problem transfers directly to
15 diversity-aware and continuously evolving safety benchmarks wherever coverage
16 is scored from pairwise (dis)agreement rates, and we describe the calibration as a
17 drop-in diagnostic rather than a replacement for search-based diversity methods.

18 1 The measurement problem

19 A recurring move in both classifier-ensemble diversity and, increasingly, in diversity- and coverage-
20 aware safety evaluation is to summarize a panel’s behavioral spread from pairwise (dis)agreement:
21 how often do two members fail, or answer, the same way. This workshop’s own scope lists “diversity
22 and coverage metrics for safety evaluation” and “behavioral repertoires of failures” as central objects
23 of study. We show that the most common member of this family — mean pairwise co-failure, one
24 of the ten classifier-diversity measures catalogued by Kuncheva & Whitaker (2003) under the name
25 double-fault — has a property that makes it a poor diversity signal on its own: it is a deterministic
26 function of item-level difficulty alone.

27 **Lemma 1 (Mean co-failure depends only on item margins)** *Let $F_{im} = 1$ iff panel member i fails*
28 *item m , with c_m the number of members failing item m . Then $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$, a*
29 *function of the column counts $\{c_m\}$ alone.*

30 Consequently, any randomization of which member fails which item that preserves each item’s failure
31 count leaves O exactly invariant, with zero variance. A benchmark with a few very hard items will

Benchmark	N	PR obs. (calibrated)	PR / null	rms excess ($\times$ null)	latent dims
ARC-Challenge	1362	23.6	0.052	$5.5\times$	6
Winogrande	1361	20.1	0.041	$4.6\times$	5
TruthfulQA	1334	17.4	0.056	$4.1\times$	5
GSM8K	1228	118.6	0.214	$2.8\times$	4
HellaSwag	1362	27.3	0.028	$11.8\times$	12

Table 1: Participation ratio (PR) of the residual correlation spectrum after exact-margin conditioning, against its own exact-margin null (mean of 400 curveball replicates). “Latent dims” counts eigenvalues exceeding the null’s spectral edge (parallel analysis against the exact null rather than iid noise). Before conditioning, the same statistic reads 1.5–3.6 on every benchmark, driven almost entirely by the shared item-difficulty eigenvalue.

show high mean co-failure even if every member fails independently given the item; a benchmark with uniformly moderate items will show low co-failure even if members are identical clones. The statistic conflates *item difficulty* with *shared behavior*, and no amount of more data resolves this — it is an identity, not a sampling artifact.

2 A calibrated diversity signal, at scale

The fix is to condition on the item margins exactly rather than compare against an uninformative uniform baseline. The row and column margins of a binary outcome matrix are the jointly sufficient statistics of the Rasch model, so the uniform distribution over margin-matched matrices — sampled here with curveball (Strona et al., 2014), verified against direct enumeration on small fibres — is the unique exact, fit-free null that removes item difficulty without fitting anything. We apply this to the archived Open LLM Leaderboard: 1,228–1,362 open models per benchmark, five benchmarks, harvested from result files at zero inference cost.

Table 1 makes the calibration gap concrete. The uncalibrated participation ratio — the same summary statistic, computed the naive way — sits at 1.5–3.6 on every benchmark: read directly, that looks like near-total redundancy in the panel. After conditioning, real structure appears: the calibrated ratio is only 2.8–21.4% of its own null, meaning the panel retains substantially more residual, non-difficulty-driven correlated structure than the calibrated comparison would suggest by coincidence, and that structure is not one dominant shared factor but 4–12 separate latent dimensions (eigenvalues above the null’s spectral edge). Neither number is visible from the mean-level statistic alone.

3 This is not just duplicate members

A natural objection to any diversity shortfall is that it is driven by near-duplicate panel members. We check this directly: clustering models at agreement 1.0000 and keeping one representative per cluster removes 900 of ARC’s 1,362 models, and moves the calibrated ratio from 0.052 only to 0.058. We also check whether ordinary discrimination heterogeneity among otherwise-independent members is sufficient to produce the effect: a two-parameter item-response simulation with realistic discrimination variance tops out at rms residual correlation 0.076, against an observed 0.248 on ARC. What best matches the observed spectral signature ($\lambda_1^2 / \sum \lambda^2 = 0.540$) is a simulated population with two *additional* latent ability dimensions (0.457), not a population of duplicated members (0.102). The calibrated diagnostic is therefore picking up genuine multi-dimensional behavioral structure, not an artifact of redundant submissions or of a single unmodeled discrimination parameter.

4 Relevance to diversity-aware benchmark governance

This workshop’s scope explicitly includes diversity-aware and continuously evolving safety benchmarks and coverage metrics for dynamic evaluation. The measurement failure mode we describe is not specific to static multiple-choice accuracy: any coverage or diversity score computed from raw pairwise (dis)agreement rates over a set of test cases — attack variants, red-team probes, behavioral test suites — inherits the same degeneracy whenever test-case difficulty (or, in a safety setting, attack potency) varies across the suite. A search procedure that appears to be discovering diverse failure

69 modes because raw disagreement is high may instead be finding items of heterogeneous difficulty;
70 a procedure that looks converged because raw agreement is high may be sitting on a benchmark
71 of uniform difficulty despite covering genuinely distinct behavioral axes. Margin-calibration —
72 comparing against the exact conditional null implied by the test suite’s own per-item difficulty profile,
73 rather than against an uninformative uniform baseline — is a domain-agnostic correction available
74 as a diagnostic wherever a diversity or coverage score is built from pairwise outcome comparisons,
75 independent of whether the underlying search procedure is evolutionary, adversarial, or manual.

76 **5 Limitations**

77 We study existing open models on static, single-turn multiple-choice benchmarks, not the multi-turn,
78 tool-using, or agentic settings central to this workshop; we have not applied this calibration inside an
79 active evolutionary or quality-diversity search loop, and doing so is the natural next step rather than
80 something we report here. Our conditioning model is unidimensional (Rasch); Section 3’s diagnostics
81 rule out discrimination heterogeneity and duplicate members as sufficient explanations but cannot
82 fully separate genuine behavioral concentration from an underfit null family. Claims are restricted to
83 the evaluated open-model ecosystem. Code, data, and the pre-registration are archived and linked in
84 the full paper.

85 **References**

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